

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay_compression.
- Best CVRPLIB holdout gap: cvrp_failure_replay_compression.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_194852_p
- started_at_local: 2026-05-13 19:48:52
- finished_at_local: 2026-05-13 19:52:09
- duration_hhmm: 00:03
- duration_seconds: 197.126
- seed_offset: 15000
- replicate_label: p
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.250831	0.012431	0.144875	0.823728	0.58	-0.004399
cvrp_random_replay	random	score_only	False	0.263081	0.005376	0.148545	0.912913	0.58	0.001818
cvrp_failure_replay	failure	score_only	False	0.261546	0.005376	0.147693	0.74887	0.58	-0.007923
cvrp_failure_replay_compression	failure	novelty_gate	True	0.245399	0.005376	0.138722	0.0	0.58	0.0

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.250831, synthetic 0.012431, combined 0.144875.

- Accepted-epoch count 3, mean accepted code novelty 0.823728, and final complexity 0.58.
- Adaptation efficiency -0.004399 and archive sizes {'worst_cases': 5, 'failure_cases': 4, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.250831 across 5 instances; family means: A=0.103839, B=0.272417, E=0.285988, P=0.319495.
- Panel synthetic_holdout mean gap 0.012431 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.028221.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 581, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.268559}, {"best_known_cost": 937, "cost": 1062, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.133404}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.263081, synthetic 0.005376, combined 0.148545.
- Accepted-epoch count 2, mean accepted code novelty 0.912913, and final complexity 0.58.
- Adaptation efficiency 0.001818 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.263081 across 5 instances; family means: A=0.115183, B=0.272417, E=0.335893, P=0.319495.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 587, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.281659}, {"best_known_cost": 937, "cost": 1062, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.133404}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.261546, synthetic 0.005376, combined 0.147693.
- Accepted-epoch count 2, mean accepted code novelty 0.74887, and final complexity 0.58.
- Adaptation efficiency -0.007923 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.261546 across 5 instances; family means: A=0.143979, B=0.272417, E=0.299424, P=0.319495.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 591, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.290393}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 858, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.094388}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.

- Final transfer gaps: CVRPLIB 0.245399, synthetic 0.005376, combined 0.138722.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.245399 across 5 instances; family means: A=0.143979, B=0.248809, E=0.28215, P=0.303249.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 591, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.290393}, {"best_known_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.145144}, {"best_known_cost": 784, "cost": 858, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.094388}]

Judge Appendix

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Summary of Replay-Aware CVRP Benchmark Suite Results

Condition	Replay Mode	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Accepted Epochs	Adaptation Efficiency	Mean Code Novelty	Compression Pressure	Notes on Transfer & Optimality
cvrp_no_replay	none	0.2508	0.0124	0.1449	3	-0.0044	0.824	No	Baseline transfer moderate; moderate CVRPLIB gap
cvrp_random_replay	random	0.2631	**0.0054**	0.1485	2	0.0018	0.913	No	Best synthetic holdout gap (lowest), slightly worse CVRPLIB gap vs no_replay, transfer gap slightly worse
cvrp_failure_replay	failure	0.2615	0.0054	0.1477	2	-0.0079	0.749	No	Similar synthetic and transfer gaps as random replay; CVRPLIB gap close to random replay
cvrp_failure_replay_compression	failure + compression-aware	**0.2454**	0.0054	**0.1387**	1	0.0	0.0	Yes	Best CVRPLIB gap and transfer gap, lowest code novelty (zero), compression active

Interpretation

- **Main Evidence for Transfer: CVRPLIB & Synthetic Holdout Gaps**
- **Best transfer** (lowest mean transfer gap) is achieved by **cvrp_failure_replay_compression** (0.1387), also the best on held-out CVRPLIB (0.2454).
- **Synthetic holdout gap** is consistently lowest (0.0054) for replay conditions: random replay, failure replay, and failure replay with compression. No replay condition has a slightly higher synthetic gap (0.0124).
- Random replay has marginally higher transfer gap (0.1485) and CVRPLIB gap (0.2631) compared to failure replay and failure replay compression.
- **Code Novelty vs Transfer**

- Code novelty ****drops substantially to zero**** in the compression-aware failure replay condition, while transfer metrics improve.
- Random replay shows the highest code novelty (~0.91) but slightly worse transfer.
- Thus, ****code novelty decreases while transfer improves**** when switching to compression-aware failure replay, indicating improvements are likely not driven by lexical novelty but by replay + compression mechanisms.
- ****Replay Mode Differences****
- No replay shows the poorest transfer and held-out CVRPLIB performance.
- Random replay improves synthetic holdout gap significantly but does not improve CVRPLIB gap compared to no replay.
- Failure replay alone improves synthetic gap but not noticeably better than random replay on CVRPLIB gap.
- Failure replay with compression yields the best transfer and CVRPLIB gaps, despite lower acceptance (only 1 accepted epoch) and zero code novelty.
- ****Complexity and Behavior Profile****
- Final complexity and behavior profiles are constant across conditions (0.58 and balanced).
- Archive sizes and epoch counts are similar, except accepted epochs drop to 1 when compression applied.

Conservative Conclusions

- ****cvrp_failure_replay_compression condition objectively achieves the best transfer performance****, as evidenced by the lowest held-out CVRPLIB gap (0.2454) and lowest transfer gap (0.1387).
- This condition reduces code novelty to zero, indicating transfer improvements occur through replay + compression mechanisms rather than exploratory code novelty.
- Random replay produces the best synthetic holdout optimality gap (0.0054) but does not match the best CVRPLIB or transfer gap.
- No replay condition underperforms others on key transfer metrics.
- Compression-aware failure replay is the preferred replay strategy for improving ultimate CVRP generalization to held-out benchmarks.
- Optimality gaps remain sizeable on held-out CVRPLIB instances (all >10%), leaving room for further improvement.

Analysis prioritized deterministic metrics (gaps, transfer) and considered replay modes distinctly. Lexical novelty relevance was contextualized against transfer results.